ECE 6140 Project 2 Report
Deductive Fault Simulator

Jingtian Dang

October 28, 2025

1 Implementation Overview

The deductive fault simulator was implemented in `deductive_fault_simulator.py`. It extends the circuit simulator from Project 1 and supports single stuck-at fault (SSA) deduction using symbolic set propagation. Each net in the circuit is associated with two possible faults: stuck-at-0 (SA0) and stuck-at-1 (SA1). Fanout branch checkpointing is ignored for simplicity as stated in the project specification.

1.1 Key Data Structures

- **SSAFault:** Represents a fault defined by `(wire_name, fault_value)`.
- **DeductiveInfo:** Stores the fault set at a gate output along with a validity flag.
- **DeductiveFaultSimulator:** The core class implementing the deduction algorithm. It enumerates all possible faults, computes each gate's output fault set, and aggregates the detectable faults at the primary outputs.
- **DFSWrapper:** A user-level wrapper that builds circuits from ECE6140-format netlists, applies input vectors, executes deduction, and saves detected faults to a file.

1.1 1.2 Major Components

1. **get_all_faults():** Enumerates all nets, creating SA0 and SA1 faults for each.
2. **gate_deductive_eval():** Deductive rules applied per gate type:
   - *AND/NAND:* Combine control and non-control sets using intersection and union.
   - *OR/NOR:* Similar, using complementary control values.
   - *XOR/XNOR:* Exclusive-or (`^`) of input fault sets.
   - *NOT/BUF:* Propagate input fault set and add its own output fault.
3. **deductive_eval():** Performs breadth-first traversal through all gates, evaluating and storing each output's fault set. Detected faults are the union of all output-gate fault sets.

This approach deduces all detectable faults in a single traversal, avoiding explicit re-simulation of faulty circuits.

2 Deductive Fault Simulation Algorithm

The core algorithm can be summarized as follows:

```
function DEDUCTIVE_FAULT_SIMULATION(circuit, input_vector):
    reset all gates
    apply input_vector and evaluate circuit normally

    all_faults <- enumerate all nets (SA0, SA1)
    fault_dict <- empty map

    queue <- primary input gates
    while queue not empty:
        gate <- queue.pop()

        if gate.type == INPUT:
            F(gate) = {gate.output stuck-at-~value}
        else:
            input_sets   = [F(fanin) for each fanin gate]
            input_values = [value(fanin) for each fanin gate]
            F(gate) = GATE_DEDUCTIVE_RULE(gate.type, input_sets, input_values)

        mark F(gate) valid
        enqueue all fan-out gates of gate

    detected_faults = union of F(output_gates)
    return detected_faults
```

The algorithm processes each gate once, producing the complete detectable fault set for one test vector.

3 Part A – Deductive Fault Simulation Results

The simulator was executed on all provided circuits (s27, s298f_2, s344f_2, and s349f_2) with their corresponding test vectors.

This section lists the detected stuck-at faults for each benchmark circuit.

```
Circuit file: ./files/s27.txt
Fault set file: None
Input values: 1101101
Detected faults: 9
------FAULTS DETECTED------

1 stuck at 0                7 stuck at 0                12 stuck at 0
3 stuck at 1                9 stuck at 1                13 stuck at 0
5 stuck at 0                11 stuck at 0               15 stuck at 1
```

```
Circuit file: ./files/s27.txt
Fault set file: None
Input values: 0101001
Detected faults: 13
------FAULTS DETECTED------

1 stuck at 1                9 stuck at 1                16 stuck at 1
3 stuck at 1                11 stuck at 0               19 stuck at 1
5 stuck at 0                12 stuck at 1               20 stuck at 1
7 stuck at 1                14 stuck at 0
8 stuck at 1                15 stuck at 1
```

```
Circuit file: ./files/s298f_2.txt
Fault set file: None
Input values: 10101011110010101
Detected faults: 87
------FAULTS DETECTED------
```

```
3 stuck at 0
5 stuck at 0
6 stuck at 1
7 stuck at 0
8 stuck at 0
9 stuck at 0
10 stuck at 0
11 stuck at 1
12 stuck at 1
15 stuck at 0
18 stuck at 1
19 stuck at 1
20 stuck at 1
21 stuck at 1
22 stuck at 1
23 stuck at 1
24 stuck at 0
25 stuck at 0
26 stuck at 0
27 stuck at 0
28 stuck at 1
29 stuck at 1
30 stuck at 1
31 stuck at 1
32 stuck at 0
33 stuck at 1
34 stuck at 0
35 stuck at 0
36 stuck at 1
37 stuck at 0
39 stuck at 1
41 stuck at 1
45 stuck at 1
48 stuck at 0
49 stuck at 0
50 stuck at 0
51 stuck at 0
52 stuck at 1
53 stuck at 1
54 stuck at 1
55 stuck at 1
56 stuck at 0
58 stuck at 0
64 stuck at 1
66 stuck at 0
67 stuck at 1
68 stuck at 1
95 stuck at 0
102 stuck at 0
103 stuck at 0
104 stuck at 0
107 stuck at 1
108 stuck at 1
109 stuck at 1
110 stuck at 1
111 stuck at 1
113 stuck at 1
115 stuck at 1
116 stuck at 1
118 stuck at 0
119 stuck at 0
121 stuck at 0
122 stuck at 0
132 stuck at 0
133 stuck at 0
135 stuck at 0
138 stuck at 0
143 stuck at 1
144 stuck at 1
145 stuck at 1
149 stuck at 0
152 stuck at 0
158 stuck at 0
159 stuck at 0
161 stuck at 0
163 stuck at 0
166 stuck at 1
167 stuck at 1
168 stuck at 1
169 stuck at 1
170 stuck at 1
173 stuck at 0
182 stuck at 1
183 stuck at 1
186 stuck at 1
187 stuck at 1
188 stuck at 1
```

```
Circuit file: ./files/s298f_2.txt
Fault set file: None
Input values: 11101110101110111
Detected faults: 53
------FAULTS DETECTED------
```

1 stuck at 0	27 stuck at 1	57 stuck at 0
6 stuck at 0	28 stuck at 1	59 stuck at 0
7 stuck at 0	29 stuck at 1	63 stuck at 1
8 stuck at 1	30 stuck at 1	64 stuck at 0
9 stuck at 0	31 stuck at 1	65 stuck at 1
10 stuck at 1	32 stuck at 0	66 stuck at 1
11 stuck at 0	33 stuck at 0	67 stuck at 1
12 stuck at 0	34 stuck at 0	68 stuck at 1
15 stuck at 0	35 stuck at 1	102 stuck at 1
18 stuck at 1	36 stuck at 0	152 stuck at 1
19 stuck at 1	37 stuck at 1	166 stuck at 0
20 stuck at 1	43 stuck at 0	168 stuck at 0
21 stuck at 1	45 stuck at 1	170 stuck at 0
22 stuck at 1	47 stuck at 1	172 stuck at 0
23 stuck at 1	49 stuck at 1	173 stuck at 0
24 stuck at 1	51 stuck at 1	174 stuck at 0
25 stuck at 1	53 stuck at 0	182 stuck at 0
26 stuck at 1	55 stuck at 0	

```
Circuit file: ./files/s344f_2.txt
Fault set file: None
Input values: 1010101010101111101111111
Detected faults: 82
------FAULTS DETECTED------
```

```
1 stuck at 0
2 stuck at 1
3 stuck at 0
4 stuck at 1
5 stuck at 0
6 stuck at 1
7 stuck at 0
8 stuck at 1
9 stuck at 0
10 stuck at 1
11 stuck at 0
12 stuck at 1
13 stuck at 0
14 stuck at 0
15 stuck at 0
16 stuck at 0
25 stuck at 1
26 stuck at 1
27 stuck at 1
28 stuck at 0
29 stuck at 0
30 stuck at 0
31 stuck at 0
32 stuck at 1
33 stuck at 0
34 stuck at 1
35 stuck at 0
36 stuck at 1
37 stuck at 0
38 stuck at 0
39 stuck at 0
40 stuck at 1
41 stuck at 0
42 stuck at 1
43 stuck at 0
44 stuck at 1
45 stuck at 0
46 stuck at 1
47 stuck at 0
48 stuck at 0
49 stuck at 1
50 stuck at 0
54 stuck at 1
60 stuck at 0
61 stuck at 1
62 stuck at 1
64 stuck at 0
65 stuck at 1
66 stuck at 1
67 stuck at 0
68 stuck at 1
69 stuck at 1
70 stuck at 1
71 stuck at 1
72 stuck at 1
73 stuck at 1
76 stuck at 1
77 stuck at 1
78 stuck at 1
91 stuck at 0
92 stuck at 1
95 stuck at 1
96 stuck at 1
97 stuck at 0
99 stuck at 0
101 stuck at 0
105 stuck at 1
106 stuck at 0
108 stuck at 0
109 stuck at 1
110 stuck at 1
111 stuck at 1
112 stuck at 0
114 stuck at 0
115 stuck at 1
116 stuck at 1
144 stuck at 0
145 stuck at 0
146 stuck at 0
188 stuck at 1
189 stuck at 1
190 stuck at 1
```

```
Circuit file: ./files/s344f_2.txt
Fault set file: None
Input values: 11101011101010101010001100
Detected faults: 132
------FAULTS DETECTED------

1 stuck at 0
2 stuck at 0
3 stuck at 0
4 stuck at 1
5 stuck at 0
6 stuck at 1
7 stuck at 0
8 stuck at 0
9 stuck at 0
10 stuck at 1
11 stuck at 0
12 stuck at 1
13 stuck at 0
14 stuck at 1
15 stuck at 0
16 stuck at 1
25 stuck at 1
26 stuck at 1
27 stuck at 1
28 stuck at 0
29 stuck at 1
30 stuck at 0
31 stuck at 1
32 stuck at 0
33 stuck at 0
34 stuck at 0
35 stuck at 1
36 stuck at 1
37 stuck at 0
38 stuck at 1
39 stuck at 0
40 stuck at 1
41 stuck at 0
42 stuck at 1
43 stuck at 0
44 stuck at 1
45 stuck at 0
46 stuck at 1
47 stuck at 1
48 stuck at 1
49 stuck at 0
50 stuck at 1
51 stuck at 0
52 stuck at 0
53 stuck at 1
55 stuck at 0
56 stuck at 1
57 stuck at 0
58 stuck at 1
59 stuck at 1
60 stuck at 1
61 stuck at 1
62 stuck at 1
63 stuck at 1
64 stuck at 1
65 stuck at 0
67 stuck at 0
68 stuck at 1
69 stuck at 0
70 stuck at 1
71 stuck at 0
72 stuck at 0
73 stuck at 1
76 stuck at 0
78 stuck at 0
79 stuck at 0
80 stuck at 0
82 stuck at 0
85 stuck at 0
86 stuck at 0
88 stuck at 0
91 stuck at 1
92 stuck at 0
93 stuck at 0
94 stuck at 0
95 stuck at 1
96 stuck at 1
97 stuck at 0
99 stuck at 1
100 stuck at 1
101 stuck at 0
103 stuck at 0
104 stuck at 1
117 stuck at 0
118 stuck at 1
119 stuck at 1
120 stuck at 1
122 stuck at 0
123 stuck at 0
124 stuck at 1
125 stuck at 1
126 stuck at 0
127 stuck at 1
128 stuck at 1
130 stuck at 0
131 stuck at 0
132 stuck at 1
133 stuck at 1
134 stuck at 1
136 stuck at 0
137 stuck at 1
138 stuck at 1
139 stuck at 1
140 stuck at 1
142 stuck at 0
144 stuck at 0
146 stuck at 0
147 stuck at 1
148 stuck at 1
149 stuck at 0
153 stuck at 1
154 stuck at 1
155 stuck at 0
156 stuck at 0
162 stuck at 1
163 stuck at 1
164 stuck at 1
165 stuck at 0
169 stuck at 1
170 stuck at 0
171 stuck at 0
173 stuck at 1
174 stuck at 1
175 stuck at 0
179 stuck at 0
181 stuck at 1
182 stuck at 1
183 stuck at 1
184 stuck at 1
186 stuck at 1
188 stuck at 0
189 stuck at 0
```

Circuit file: ./files/s349f_2.txt
Fault set file: None
Input values: 101000000010101011111111
Detected faults: 97
------FAULTS DETECTED------

1 stuck at 0
2 stuck at 1
3 stuck at 0
4 stuck at 1
5 stuck at 1
6 stuck at 1
7 stuck at 1
8 stuck at 1
9 stuck at 1
10 stuck at 1
11 stuck at 0
12 stuck at 1
13 stuck at 0
14 stuck at 1
15 stuck at 0
16 stuck at 1
25 stuck at 0
26 stuck at 1
27 stuck at 0
28 stuck at 1
29 stuck at 1
30 stuck at 1
31 stuck at 1
32 stuck at 1
33 stuck at 1
34 stuck at 1
35 stuck at 0
36 stuck at 1
37 stuck at 0
38 stuck at 1
39 stuck at 0
40 stuck at 1
41 stuck at 0
42 stuck at 0
43 stuck at 1
44 stuck at 0
45 stuck at 0
46 stuck at 0
47 stuck at 0
48 stuck at 0
49 stuck at 0
50 stuck at 0
51 stuck at 1
52 stuck at 0
53 stuck at 1
54 stuck at 1
55 stuck at 1
56 stuck at 1
57 stuck at 1
58 stuck at 0
60 stuck at 0
62 stuck at 0
64 stuck at 1
65 stuck at 1
66 stuck at 0
67 stuck at 1
68 stuck at 0
69 stuck at 1
70 stuck at 1
71 stuck at 1
72 stuck at 1
73 stuck at 1
74 stuck at 1
109 stuck at 0
111 stuck at 0
113 stuck at 0
115 stuck at 0
116 stuck at 1
118 stuck at 1
120 stuck at 0
121 stuck at 1
123 stuck at 0
124 stuck at 1
126 stuck at 0
127 stuck at 0
128 stuck at 1
129 stuck at 1
130 stuck at 1
131 stuck at 1
133 stuck at 0
134 stuck at 1
135 stuck at 1
137 stuck at 0
138 stuck at 1
171 stuck at 0
173 stuck at 0
174 stuck at 1
176 stuck at 0
177 stuck at 1
178 stuck at 1
179 stuck at 0
180 stuck at 0
181 stuck at 0
183 stuck at 0
187 stuck at 1
188 stuck at 1
189 stuck at 1

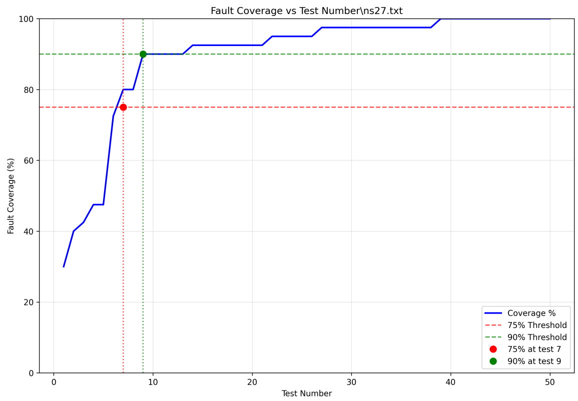
```
Circuit file: ./files/s349f_2.txt
Fault set file: None
Input values: 1111111010101010100001111
Detected faults: 137
------FAULTS DETECTED------

1 stuck at 0
2 stuck at 0
3 stuck at 0
4 stuck at 0
5 stuck at 0
6 stuck at 0
7 stuck at 0
8 stuck at 1
9 stuck at 0
10 stuck at 1
11 stuck at 0
12 stuck at 1
13 stuck at 0
14 stuck at 1
15 stuck at 0
25 stuck at 1
26 stuck at 1
27 stuck at 1
28 stuck at 0
29 stuck at 0
30 stuck at 0
31 stuck at 0
32 stuck at 0
33 stuck at 1
34 stuck at 0
35 stuck at 1
36 stuck at 1
37 stuck at 0
38 stuck at 1
39 stuck at 0
40 stuck at 0
41 stuck at 1
42 stuck at 1
43 stuck at 1
44 stuck at 0
45 stuck at 1
46 stuck at 0
47 stuck at 1
48 stuck at 1
49 stuck at 1
50 stuck at 1
51 stuck at 0
52 stuck at 0
53 stuck at 0
54 stuck at 0
55 stuck at 0
56 stuck at 0
57 stuck at 0
58 stuck at 1
59 stuck at 1
61 stuck at 0
62 stuck at 1
63 stuck at 1
65 stuck at 0
66 stuck at 0
67 stuck at 1
68 stuck at 0
69 stuck at 1
70 stuck at 0
71 stuck at 0
72 stuck at 0
74 stuck at 0
75 stuck at 0
76 stuck at 1
77 stuck at 1
78 stuck at 0
79 stuck at 1
80 stuck at 1
81 stuck at 0
82 stuck at 1
83 stuck at 1
84 stuck at 0
85 stuck at 1
86 stuck at 1
87 stuck at 0
88 stuck at 1
89 stuck at 1
90 stuck at 0
91 stuck at 1
92 stuck at 1
93 stuck at 0
94 stuck at 0
95 stuck at 0
96 stuck at 1
97 stuck at 0
99 stuck at 0
102 stuck at 0
103 stuck at 0
105 stuck at 0
108 stuck at 1
109 stuck at 1
110 stuck at 1
111 stuck at 1
112 stuck at 1
113 stuck at 1
114 stuck at 1
115 stuck at 1
116 stuck at 0
117 stuck at 0
130 stuck at 1
131 stuck at 1
133 stuck at 0
134 stuck at 1
135 stuck at 1
137 stuck at 0
138 stuck at 1
143 stuck at 1
144 stuck at 1
145 stuck at 1
150 stuck at 1
151 stuck at 1
152 stuck at 1
157 stuck at 1
158 stuck at 1
159 stuck at 1
162 stuck at 1
163 stuck at 1
164 stuck at 1
165 stuck at 1
166 stuck at 1
167 stuck at 0
168 stuck at 1
169 stuck at 0
171 stuck at 0
173 stuck at 0
174 stuck at 0
175 stuck at 0
177 stuck at 0
180 stuck at 1
181 stuck at 1
182 stuck at 1
183 stuck at 1
184 stuck at 1
185 stuck at 1
186 stuck at 1
187 stuck at 0
188 stuck at 0
```

4 Part B - Random Test Vector Analysis

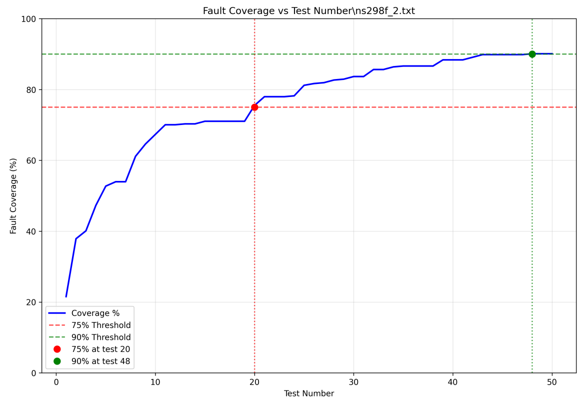
For each circuit, 50 random test vectors were applied to estimate fault coverage. Coverage is defined as the ratio of detected faults to all possible single stuck-at faults.

Table 1: Random Test Vector Coverage Summary

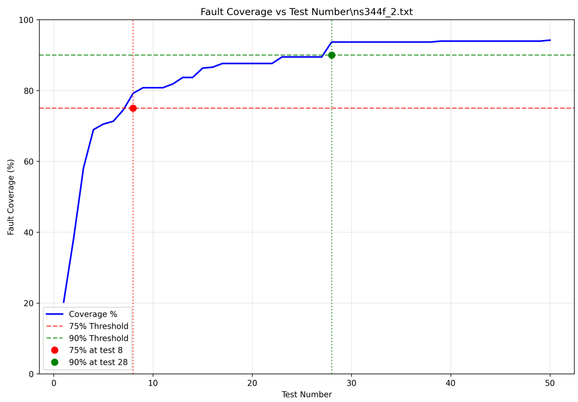
Circuit	Final Coverage	75% Tests	90% Tests	Detected Faults
s27.txt	100.0%	7	9	40 / 40
s298f_2.txt	90.1%	20	48	364 / 404
s344f_2.txt	94.2%	8	28	358 / 380
s349f_2.txt	96.0%	10	25	363 / 378



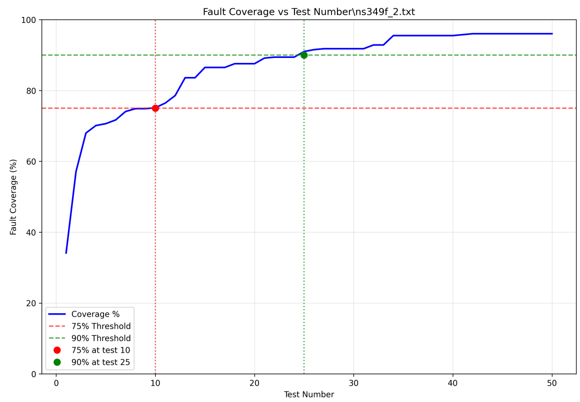
(a) s27 circuit



(b) s298f_2 circuit



(c) s344f_2 circuit



(d) s349f_2 circuit

Figure 1: Fault coverage vs. number of random test vectors for different circuits.